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Lanostane triterpenoids from *Ganoderma hainanense* J. D. Zhao

XingRong Peng [a,b], JieQing Liu [a], JianJun Xia [a], CuiFang Wang [a], XuYang Li [a,b], YuanYuan Deng [a,b], NiMan Bao [a,b], ZhiRun Zhang [a], MingHua Qiu [a,*]

[a] State Key Laboratory of Phytochemistry and Plant Resources in West China, Kunming Institute of Botany, Chinese Academy of Sciences, Kunming 650201, People's Republic of China
[b] Graduate University of the Chinese Academy of Sciences, Beijing 100049, People's Republic China

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ABSTRACT

Chemical investigation of the fruiting bodies of *Ganoderma hainanense* resulted in isolation of fourteen lanostane triterpenoids, including nine ganoderma acids and five ganoderma alcohols, together with five known compounds. Structural elucidation was determined using extensive spectroscopic technologies, Mosher's method and X-ray single crystal diffraction. Three of the compounds showed inhibitory activities against HL-60, SMMC-7721, A-549 and MCF-7 cells with IC_{50} values of 15.0–40.0 μM.

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1. Introduction

Ganoderma species have long been used for treating and preventing various diseases in Oriental countries due to their beneficial effects (Sanodiya et al., 2009). Lanostane triterpenes are one of the main components, and have some useful biological activities, including cytotoxic (lucidenic acid A, lucidenic acid A and ganoderic acid E) (Wu et al., 2001), farnesoid X receptor (FXR)-inducing (lucidumol A, ganoderic acid TR, ganodermanontriol and ganoderiol F) (Grienke et al., 2011), aldose reductase inhibitory (ganoderic acid Df, ganoderic acid C1, ganoderic acid C2 and ganoderenic acid A) (Fatmawati et al., 2011a), 5α-reductase inhibitory (ganoderic acid DM) (Liu et al., 2006), anti-HIV protease (lucidumol A and ganoderic acid β) (Min et al., 1998), anti-inflammatory (ganoderic acids A, C1, C2 and DM) (Akihisa et al., 2007), and α-glucosidase inhibitory (ganoderol B) properties (Fatmawati et al., 2011a), respectively.

Of these, *Ganoderma hainanense*, a rare species of *Ganoderma*, has traditionally been used for immuno-regulation, liver-protection, anti-aging, preventing cardiovascular disease, treating hypertension and diabetes mellitus (Li et al., 2013). However, only one paper has reported isolation and identification of a single new lanostane triterpenoid from this fungus (Ma et al., 2013).

Since 2004, the research focus has been on bioactive chemical constituents from the genus *Ganoderma* (i.e. *G. fornicatum*, *G. sinense* and *G. resinaceum*) (Niu et al., 2004; Wang et al., 2010; Liu et al., 2012a; Peng et al., 2013). In continuing work, fruiting bodies of *G. hainanense* were systematically researched and nineteen lanostane triterpenoids were isolated (Fig. 1). The structures of fourteen new isolates were elucidated by extensive spectroscopic analysis, and use of Mosher's method and X-ray crystallography, respectively. The cytotoxicities of compounds **1**, **3**–**8**, **10**–**14**, **16**–**19** were also evaluated against HL-60, SMMC-7721, A-549, MCF-7 and SW380 cell lines by the MTT assay.

2. Results and discussion

2.1. Structures of new compounds

The methanol extract of *G. hainanense* was subjected to chromatographic purification using Sephadex LH-20 and repeated silica gel column chromatographic steps, followed by preparative HPLC, to afford fourteen new compounds **1**–**14**. In addition, five known compounds, ganoderol J (**15**) (Liu et al., 2012b), ganoderone A (**16**) (Niedermeyer et al., 2005), lucidadiol (**17**) (Gonzalez et al., 1999), ganodermanontriol (**18**) (Nishitoba et al., 1988), and 4,4,14α-trimethyl-3,7-dioxo-5α-chol-8-en-24-oic acid (**19**) (Cheng et al., 2010) were isolated and identified by comparison of their spectroscopic data from the literature.

Ganohainanic acid A (**1**) gave a molecular ion peak at *m*/*z* 530.2885 $[M]^+$ (calcd for $C_{30}H_{42}O_8$, 530.2880). Its IR spectrum established presence of hydroxyl (3423 cm^{-1}) and α,β-unsaturated carbonyl (1698 cm^{-1}) groups, in agreement with the UV absorption (λ_{max} 266 and 227 nm) data. The ^1H NMR spectrum showed four singlet methyl signals at δ 1.48, 1.41, 1.34 and 1.13, two doublet methyl resonances at δ 1.04 (*J* = 6.3 Hz) and δ 1.36 (*J* = 7.2 Hz), as well as two doublet proton signals each corresponding to an

* Corresponding author. Tel.: +86 0871 5223327; fax: +86 0871 5223325.
E-mail address: mhchiu@mail.kib.ac.cn (M. Qiu).

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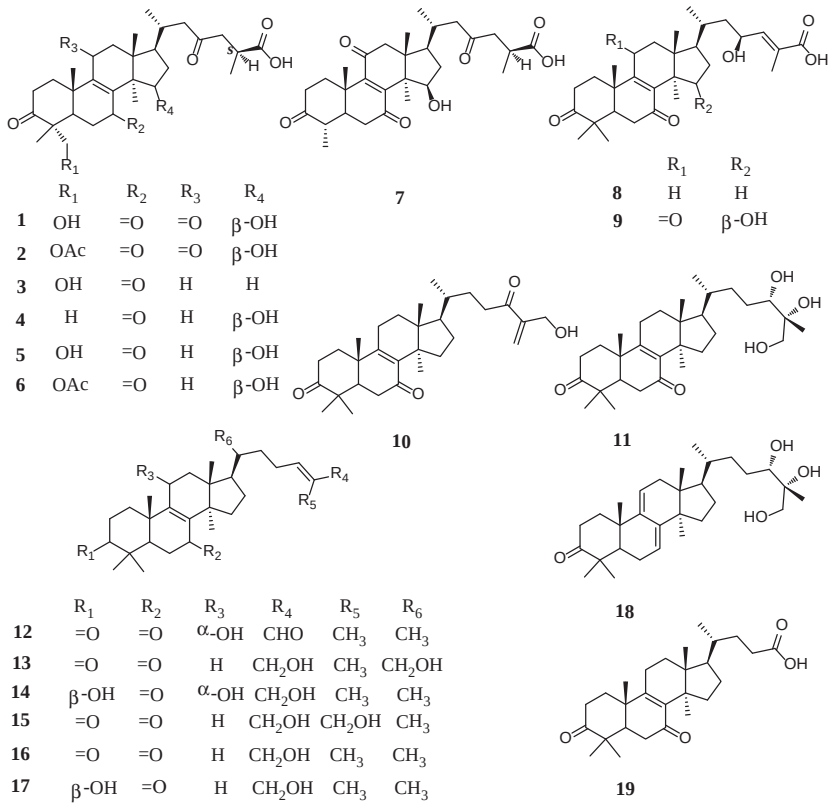

Fig. 1. Structures of compounds **1**–**19** from *G. hainanense*.

oxygenated methylene moiety at δ 3.71 (d, J = 10.5 Hz) and δ 3.84 (d, J = 10.5 Hz). The ^{13}C NMR spectroscopic data also displayed 30 carbon resonances in **1**, which were attributed to six methyls, eight methylenes (one oxygenated), four methines, and eleven quaternary carbons, including two olefinic carbons, four ketone carbonyl carbons and one carboxyl carbon. Among them, resonances at δ 203.5 (C-7), δ 151.7 (C-8), δ 150.1 (C-9) and δ 202.1 (C-11) were characteristics of a conjugated system (C-7/C-8/C-9/C-11).

The 1D-NMR spectroscopic data for **1** indicated that it was a lanostane triterpenoid with a 7,11,23-trioxo-5α-lansta-8-en-26-oic acid substructure similar to ganoderic acid J (Nishitoba et al., 1985), except that the methyl at C-28 in the latter was replaced by a hydroxymethylene in **1**. Confirmation to this interpretation was established from HMBC correlations (Fig. 2) of H_2-1, H_2-2, H-5, H_2-28 and H_3-29 with C-3 and of H_3-29 and H-5 with C-28, respectively, together with a ROESY correlation of H_2-28 with H-5. Additionally, the ROESY correlation of H-15 with H_3-30 assigned the OH-15 group as β. Methylation of **1** gave **1a**, the X-ray crystal diffraction analysis (Fig. 3) of which determined its absolute configuration at C-25 to be *S*. Thus, structure **1** was established to be 25*S*-15β,28-dihydroxy-3,7,11,23-tetraoxo-5α-lanost-8-en-26-oic acid.

Detailed comparison of the 1D-NMR spectroscopic data between **2** (HREIMS, m/z 572.2978 [M]$^+$, calcd for $C_{32}H_{44}O_9$, 572.2985) and **1** indicated the presence of additional methyl (δ 1.89; δ 21.0) and carbonyl (δ 170.9) groups in **2**, corresponding to an acetyl moiety. Furthermore, HMBC correlations of H_2-28 with the carbonyl carbon of the acetyl group confirmed the 28-OH group of **2** was acylated. So compound **2** was 25*S*-28-acetoxyl-15β-hydroxy-3,7,11,23-tetraoxo-5α-lanost-8-en-26-oic acid, and named acetyl ganohainanic acid A.

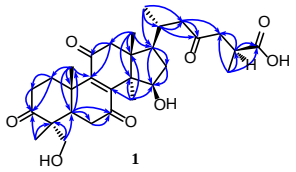

Fig. 2. Key HMBC (H → C) correlations of **1**.

A molecular formula $C_{30}H_{44}O_6$ of **3** was established from its HREIMS ([M]$^+$, m/z 500.3121, calcd 500.3138). Its 1D-NMR spectroscopic data were similar to those of **1**, except for two methylenes instead of a ketone carbonyl and an oxymethine in **1**. HMBC correlations of: H_2-16 and H-17 with C-15; H_2-15 with C-14 and C-30; H_3-18 with C-12, C-13, C-14 and C-17; and H_2-12 with C-11 and C-13 confirmed that the oxymethine at C-15 and the carbonyl group at C-11 in **1** were replaced by two methylenes in **3**, respectively. Therefore, structure **3** was 25*S*-28-hydroxy-3,7,23-trioxo-5α-lanosta-8-en-26-oic acid and named as ganohainanic acid B.

Ganohainanic acid C (**4**) and compound **3** had the same molecular formula of $C_{30}H_{44}O_6$ as determined by HREIMS (m/z 500.3140, [M]$^+$, calcd 500.3138). However, comparison of their 1D-NMR spectroscopic data showed methylene and oxymethine groups in **4** instead of the oxymethylene at C-28 and the methylene at C-15 in **3**, respectively; this interpretation was confirmed by HMBC correlations of: H_3-28 and H_3-29 with C-3 and C-4;H-5 and H_2-6 with C-4; H_3-30, H_2-16 and H-17 with C-15; and H-15 with C-14, C-16 and C-17. A ROESY correlation of H_3-30/H-15 assigned the relative configuration of the 15-OH group as β. Accordingly, structure **4** was25*S*-15β-hydroxy-3,7,23-trioxo-5α-lanost-8-en-26-oic acid, and named as ganohainanic acid C (**4**).

The molecular formula of **5** was assigned as $C_{30}H_{44}O_7$ by HREIMS ([M]$^+$, m/z 516.3039, calcd 516.3087), with one oxygen more than that of **4**. Its 1D-NMR spectroscopic data showed the presence of an oxymethylene and six methyl groups, indicating that Me-28 was replaced by a hydroxyl group in **5**. Furthermore, H_2-28 (δ 4.13, d, J = 12.0 Hz and δ 3.57, d, J = 12.0 Hz) gave HMBC correlations with C-3, C-4 and C-5, and together with the ROESY correlation

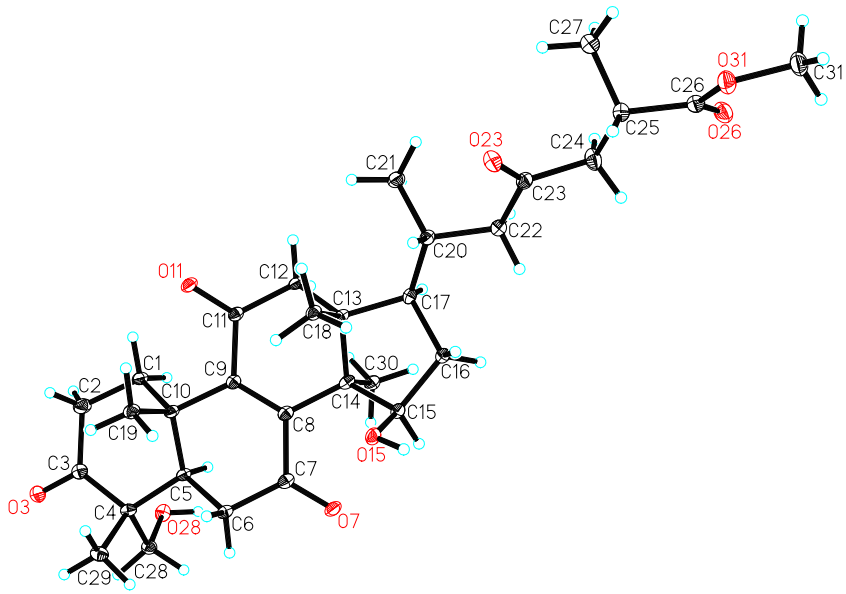

Fig. 3. The X-ray crystal structure of **1a**.

with H-5, this confirmed the above deduction. The relative configuration of 15-OH was determined to be β by a ROESY correlation of H-15/H_3-30. Thus, structure **5** was 25*S*-15β,28-dihydroxy-3,7,23-trioxo-5α-lanost-8-en-26-oic acid, and named as ganohainanic acid D.

Compound **6** had the molecular formula $C_{32}H_{46}O_8$ ([M]$^+$, *m*/*z* 558.3171) established by HREIMS. Detailed analysis of the 1D-NMR spectroscopic data (Tables 1 and 2) between **6** and **5** showed the presence of additional methyl and carbonyl moieties in **6**, indicating that the latter was likely an acetylated derivative of **5**. HMBC correlations from H_2-28 (δ 4.16, d, *J* = 10.8 Hz and δ 4.51, d, *J* = 10.8 Hz) and the methyl (δ 1.83, s) to the carbonyl carbon (δ 170.8), together with the ROESY correlation of H_2-28/H-5, confirmed that the acetyl group was located at 28-OH in **6**. Consequently, compound **6** was assigned as 25*S*-28-acetoxyl-15β-hydroxy-3,7,23-trioxo-5α-lanost-8-en-26-oic acid and named acetyl ganohainanic acid D.

Compound **7** was assigned a molecular formula of $C_{29}H_{40}O_7$ on the basis of the HREIMS (*m*/*z* 500.2769 [M]$^+$, calcd 500.2774) and NMR spectroscopic data (Tables 1 and 2). Its IR spectrum showed absorptions of hydroxyl (3441 cm^{-1}) and carbonyl (1711 and 1671 cm^{-1}) groups. Its ^{13}C NMR spectra of **7** displayed 29 carbon resonances, including six methyls, seven methylenes, six methines (one oxygenated), a conjugated system of C-7/C-8/C-9/C-11 (δ 202.7, C-7, δ 151.8, C-8, δ 151.3, C-9, δ 202.9, C-11), two ketone carbonyls (δ 210.5, C-3, δ 209.5, C-23) and one carboxyl (δ 178.8, C-26), respectively. While, these data indicated that **7** was similar to ganoderic acid J (Nishitoba et al., 1985), its NMR spectrum had displayed three doublet methyls (δ 1.38, d, *J* = 6.0 Hz; δ 1.04, d, *J* = 6.0 Hz and δ 1.03, d, *J* = 6.0 Hz) and three singlet methyls. The observed HMBC correlations of a doublet of methyl protons (δ 1.03, d, *J* = 6.0 Hz) with C-3, C-4 and C-5, and ^{1}H-^{1}H COSY correlations of methyl protons with H-4, and of H-4/H-5/H_2-6, as well as ROESY correlations of H-4/H_3-19 and of H-5/H_3-28 demonstrated that only one methyl (Me-28) was linked to C-4 (Fig. 4). The 15β-OH group was determined by the ROESY correlation of H-15/H_3-30. Thus, structure **7** was established to be 15β-hydroxy-3,7,11,23-tetraoxo-4β-H-5α-29-norlanost-8-en-26-oic acid, named ganohainanic acid E.

Hainanic acid A (**8**) was assigned a molecular formula of $C_{30}H_{44}O_5$ based on HREIMS (*m*/*z* 484.3183 [M]$^+$, calcd 484.3189) and 1D-NMR spectroscopic data (Tables 1 and 2) analysis. Its IR spectrum showed the presence of hydroxyl (3440 cm^{-1}) and α,β-unsaturated carbonyl (1709 cm^{-1}) groups, while, its UV absorption at 252 nm suggested presence of an α,β-unsaturated carbonyl group. Its ^{13}C NMR spectra had characteristic carbon signals at δ 214.5 (C), δ 198.2 (C), δ 139.7 (C) and δ 163.4 (C), attributed to a carbonyl group at C-3 and a conjugated system, similar to those of ganolucidic acid D (Nishitoba et al., 1986). Nevertheless, no obvious distinctions in the ^{13}C-NMR spectroscopic data between the conjugated system of C-7/C-8/C-9 and that of C-8/C-9/C-11 were observed. Based on HMBC correlations from H-5 and H_2-6 to the carbonyl group (δ 198.2), the conjugated system of **8** was deduced to be C-7/C-8/C-9.

The presence of a methylene group and the absence of an oxymethine carbon was also observed in ^{13}C NMR spectra of **8**, suggesting that the 15-OH group in ganolucidic acid D was absent. This interpretation was supported by HMBC correlations of H_3-30, H_2-16 and H-17 with C-15. Apart from the above deduction, the resonances of an oxymethine (67.5), a sp^2 methine (146.1), a sp^2 quaternary carbon (125.7) and a carboxyl group (171.9) in the side-chain of **8** were also similar to those of ganolucidic acid D. The absolute configuration of the side-chain moiety in **8** was then next determined by the following methods:

In its ROESY spectrum, there was no correlation of H-24/H_3-27, this indicating that the configuration of the double bond between C-24 and C-25 was *E*. The *S*-configuration of C-23 was determined by comparison of the coupling constant of $J_{H\text{-}23/H\text{-}24}$ (9.0 Hz) with ganoderic acid γ ($J_{H\text{-}23/H\text{-}24}$ = 9.2 Hz) (Min et al., 2000) and further confirmed by application of the Mosher ester procedure carried out in NMR tubes (Su et al., 2002). The ^{1}H NMR spectra of the **8r** and **8s** were obtained in NMR tubes directly. Although a strong proton signal for the excess MTPA chlorides and MTPA acids were present in the ^{1}H NMR spectra (see Supporting Information), the resonances (H-20, H_3-21, H_2-22, H-23, H-24 and H_3-27) of the diastereomeric MTPA esters (**8r** and **8s**) were clearly distinctive (see Supporting Information) with observed chemical shift differences ($\Delta\delta_{S\text{-}R}$, Fig. 5) unambiguously indicating the absolute configuration of C-23 of **8** to be *S*. Therefore, compound **8** was assigned as 23*S*-hydroxy-3,7-dioxo-5α-lanost-8,24*E*-dien-26-oic acid and named hainanic acid A.

Compound **9** was isolated as a white powder. Determination of its molecular formula ($C_{30}H_{42}O_7$) was based on HREIMS analysis. Presence of hydroxyl (3440 cm^{-1}) and α,β-unsaturated carbonyl

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Table 1
^{1}H NMR spectroscopic data (δ, C_5D_5N, 600 MHz, J in Hz) for compounds **1**–**14**.

H	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
1α	2.95, m	2.66, m	1.78, m	2.43, m	1.96, m	2.00, m	1.45, m	1.57, m	1.83, m	2.44, m	1.57, m	2.47, m	1.57, m	1.98, m
1β	2.67, m	2.81, m	1.94, m	2.73, m	1.81, m	1.70, m	3.25, m	1.89, m	3.08, m	1.58, m	1.90, m	2.81, m	1.89, m	2.52, m
2α	3.18, m	1.90, m	2.71, m	1.61, m	2.69, m	2.58, m	2.41, m	2.43, m	2.47, m	1.91, m	2.43, m	2.46, m	2.42, m	2.20, m
2β		3.22, m		1.93, m	3.34, m	2.76, m	2.55, m	2.70, m	2.68, m	2.83, m	2.71, m	2.78, m	2.71, m	2.57, m
3α	–	–	–	–	–	–	–	–	–	–	–	–	–	3.49, m
4	–	–	–	–	–	–	2.44, m	–	–	–	–	–	–	–
5α	3.38, m	2.90, m	3.29, m	2.73, m	2.83, m	2.83, m	1.97, m	2.15, m	2.47, m	2.17, m	2.17, m	2.42, m	2.17, m	2.02, m
6α	2.82, m	2.65, m	2.62, m	2.65, m	2.55, m	2.33, m	2.61, m	2.44, m	2.74, m	2.59, m	2.45, m	2.56, m	2.47, m	1.35, m
6β		2.87, m		2.54, m	1.81, m	2.76, m		2.57, m	2.50, m	2.45, m	2.60, m	2.75, m	2.60, m	2.02, m
7	–	–	–	–	–	–	–	–	–	–	–	–	–	4.80, m
11	–	–	2.24, m	2.25, m	2.29, m	2.31, m	–	2.19, m	–	2.21, m	2.20, m	4.81, m	2.24, m	–
12α	2.97, m	2.98, m	1.73, m	1.74, m	1.73, m	1.73, m	2.99, d (18.0)	1.75, m	2.98, d (16.2)	1.26, m	1.72, m	2.32, m	2.02, m	2.61, m
12β	2.79, m	2.80, m					2.81, d (18.0)		2.83, d (16.2)	1.74, m		2.63, m		2.32, m
15α	4.80, m	4.80, d (7.2)	2.41, m	4.74, m	4.73, d (6.0)	4.72, d (7.2)	4.84, d (6.0)	1.87, m	4.83, d (6.6)	2.42, m	2.44, m	2.49, m	2.01, m	2.02, m
15β			1.89, m							1.95, m	1.77, m	1.97, m	2.50, m	
16α	2.57, m	2.56, m	1.96, m	2.57, m	2.69, m	2.54, m	2.54, m	2.12, m	1.89, m	2.01, m	2.01, m	1.99, m	2.11, m	1.35, m
16β	1.77, m	1.78, m	1.31, m	1.82, m		1.81, m		1.48, m	2.73, m		1.39, m	2.07, m	1.53, m	
17α	1.81, m	1.80, m	1.52, m	1.54, m	1.51, m	1.51, m	1.79, m	1.62, m	1.92, m	1.48, m	1.53, m	1.68, m	2.17, m	1.71, m
18	1.48, s	1.47, s	0.75, s	1.24, s	1.25, s	1.25, s	1.47, s	0.69, s	1.45, s	0.72, s	0.71, s	0.82, s	0.84, s	0.80, s
19	1.41, s	1.40, s	1.22, s	1.20, s	1.21, s	1.24, s	1.52, s	1.21, s	1.32, s	1.25, s	1.24, s	1.43, s	1.24, s	1.32, s
20	2.46, m	2.47, m	1.88, m	2.47, m	2.47, m	2.46, m	2.46, m	1.68, m	1.95, m	1.45, m	1.52, m	1.45, m	1.74, s	1.59, m
21	1.04, d (6.3)	1.02, d (6.3)	1.08, d (6.0)	1.12, d (6.0)	1.11, d (6.0)	1.10, d (4.8)	1.04, d (6.0)	1.22, d (4.8)	1.18, d (4.2)	0.98, d (6.0)	1.05, d (6.0)	1.00, d (6.0)	3.99, dd (4.9, 10.8)	1.02, d (6.0)
													4.11, dd (3.3, 10.8)	
22	2.36, m	2.34, m	2.32, m	2.34, m	2.74, m	2.83, m	2.67, m	2.05, m	1.99, m	2.93, m	2.38, m	1.17, m	1.82, m	1.46, m
	2.69, m	2.68, m	2.67, m	2.19, m	2.34, m	2.70, m	2.31, m	1.86, m	1.84, m		1.90, m	1.57, m	1.77, m	1.16, m
23	–	–	–	–	–	–	–	5.08, m	5.04, s	2.83, m	2.43, m	2.21, m	2.32, m	2.25, m
										2.93, m	1.53, m	2.33, m		
													2.45, m	2.07, m
24	3.23, m	3.21, m	3.26, m	3.23, m	3.23, m	3.23, m	3.21, m	7.45, d (9.0)	7.44, m	–	4.23, d (12.0)	6.51, t (7.1)	5.85, t (6.7)	5.78, t (6.6)
	2.60, m	2.60, m	2.59, m	2.58, m	2.59, m	2.58, m	2.56, m							
25	3.34, m	3.34, m	3.36, m	3.34, m	3.34, m	3.34, m	3.34, m	–	–	–	–	–	–	–
26	–	–	–	–	–	–	–	–	–	4.86, s	4.17, d (6.0)	9.59, s	4.35, s	4.35, m
											4.35, d (6.0)			
27	1.36, d (7.2)	1.38, d (7.0)	1.40, d (6.0)	1.39, d (7.2)	1.39, d (7.2)	1.39, d (4.8)	1.38, d (6.0)	2.24, br s	2.15, br s	6.42, s 6.36, s	1.68, s	1.82, s	1.87, s	1.87, s
28	3.84, d (10.5)	4.34, d (12.0)	4.09, d (6.0)	1.09, s	4.13, d (12.0)	4.16, d (10.8)	1.03, d (6.0)	1.07, s	1.09, s	1.06, s	1.08, s	1.10, s	1.08, s	1.16, s
	3.71, d (10.5)	4.25, d (12.0)	3.58, d (6.0)		3.57, d (12.0)	4.51, d (10.8)								
29	1.13, s	1.17, s	1.06, s	1.04, s	1.03, s	1.06, s	–	1.04, s	1.06, s	1.08, s	1.06, s	1.08, s	1.05, s	1.14, s
30	1.34, s	1.35, s	1.03, s	0.97, s	0.91, s	0.91, s	1.35, s	1.10, s	1.41, s	1.09, s	1.25, s	1.54, s	1.15, s	1.57, s
CH_3CO	–	1.89, s	–	–	–	1.83, s	–	–	–	–	–	–	–	–

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Table 2
^{13}C NMR spectroscopic data (δ, 150 MHz) for compounds **1–14**.

C	1[b]	2[b]	3[b]	4[b]	5[b]	6[b]	7[b]	8[b]	9[b]	10[b]	11[a]	11[b]	12[b]	13[b]	14[b]
1	37.9, CH_2	36.2, CH_2	36.7, CH_2	35.1, CH_2	34.4, CH_2	34.5, CH_2	35.6, CH_2	35.5, CH_2	35.8, CH_2	35.7, CH_2	35.5, CH_2	35.7, CH_2	35.6, CH_2	35.7, CH_2	34.4, CH_2
2	34.4, CH_2	34.6, CH_2	34.3, CH_2	35.8, CH_2	36.1, CH_2	35.8, CH_2	38.2, CH_2	35.1, CH_2	34.8, CH_2	35.1, CH_2	34.6, CH_2	35.2, CH_2	35.8, CH_2	35.1, CH_2	35.3, CH_2
3	216.2, C	213.3, C	214.9, C	214.2, C	214.7, C	211.9, C	210.5, C	214.5, C	215.9, C	214.5, C	215.0, C	214.5, C	214.6, C	214.5, C	77.9, CH
4	53.2, C	50.9, C	52.9, C	49.1, C	53.0, C	50.7, C	44.4, CH	47.7, C	47.4, C	47.7, C	47.5, C	47.7, C	48.1, C	47.7, C	40.2, C
5	44.4, CH	46.1, CH	43.5, CH	50.9, CH	43.4, CH	44.7, CH	49.8, CH	51.0, CH	50.9, CH	50.9, CH	51.2, CH	51.0, CH	51.8, CH	51.0, CH	51.7, CH
6	38.1, CH_2	38.2, CH_2	37.9, CH_2	41.0, CH_2	39.9, CH_2	37.9, CH_2	39.8, CH_2	37.8, CH_2	38.1, CH_2	37.9, CH_2	37.4, CH_2	37.8, CH_2	38.4, CH_2	37.9, CH_2	29.1, CH_2
7	203.5, C	203.3, C	198.3, C	201.1, C	201.2, C	200.5, C	202.7, C	198.2, C	203.7, C	198.2, C	198.4, C	198.2, C	199.9, C	198.3, C	200.9, C
8	151.7, C	152.1, C	139.9, C	138.9, C	139.1, C	139.2, C	151.8, C	139.7, C	151.8, C	139.7, C	139.7, C	139.8, C	141.8, C	139.8, C	141.4, C
9	150.1, C	149.9, C	162.9, C	167.3, C	166.6, C	166.6, C	151.3, C	163.4, C	150.7, C	163.4, C	163.1, C	163.4, C	161.3, C	163.5, C	162.8, C
10	39.4, C	39.9, C	39.7, C	40.3, C	41.1, C	39.9, C	39.7, C	40.0, C	39.9, C	40.0, C	39.6, C	40.0, C	41.0, C	40.0, C	41.5, C
11	202.1, C	202.6, C	24.2, CH_2	24.7, CH_2	25.0, CH_2	24.9, CH_2	202.9, C	24.3, CH_2	202.6, C	24.3, CH_2	25.1, CH_2	24.6, CH_2	65.0, CH	24.5, CH_2	64.9, CH
12	53.2, CH_2	53.2, CH_2	30.9, CH_2	32.1, CH_2	32.6, CH_2	31.9, CH_2	53.1, CH_2	30.4, CH_2	53.2, CH_2	30.5, CH_2	30.3, CH_2	30.9, CH_2	45.9, CH_2	30.3, CH_2	46.2, CH_2
13	47.1, C	47.4, C	48.7, C	44.8, C	44.9, C	44.9, C	47.5, C	45.7, C	47.4, C	45.7, C	47.9, C	48.6, C	48.5, C	45.7, C	48.6, C
14	54.5, C	54.9, C	45.6, C	53.4, C	53.4, C	53.4, C	55.1, C	48.7, C	54.9, C	48.6, C	45.1, C	45.7, C	49.1, C	48.7, C	49.1, C
15	75.2, CH	75.0, CH	33.3, CH_2	76.9, CH	77.2, CH	76.8, CH	75.3, CH	33.1, CH_2	75.3, CH	33.0, CH_2	34.6, CH_2	34.8, CH_2	33.9, CH_2	33.2, CH_2	28.9, CH_2
16	41.2, CH_2	41.4, CH_2	29.8, CH_2	40.9, CH_2	38.5, CH_2	41.1, CH_2	40.8, CH_2	29.7, CH_2	41.8, CH_2	29.4, CH_2	29.9, CH_2	29.7, CH_2	28.8, CH_2	29.0, CH_2	29.0, CH_2
17	50.1, CH	50.2, CH	49.9, CH	50.3, CH	50.3, CH	50.2, CH	49.9, CH	50.2, CH	51.1, CH	49.9, CH	50.5, CH	50.1, CH	50.7, CH	51.0, CH	50.9, CH
18	18.4, CH_3	18.5, CH_3	16.6, CH_3	17.0, CH_3	17.0, CH_3	17.0, CH_3	18.2, CH_3	16.3, CH_3	18.5, CH_3	16.5, CH_3	16.1, CH_3	16.7, CH_3	17.8, CH_3	16.9, CH_3	17.8, CH_3
19	18.6, CH_3	18.7, CH_3	18.2, CH_3	18.4, CH_3	18.8, CH_3	18.2, CH_3	16.1, CH_3	17.9, CH_3	19.2, CH_3	18.1, CH_3	18.1, CH_3	18.2, CH_3	19.7, CH_3	18.1, CH_3	20.5, CH_3
20	33.6, CH	23.5, CH	33.6, CH	33.7, CH	33.8, CH	33.7, CH	33.5, CH_2	34.8, CH	34.9, CH	36.9, CH	37.4, CH	37.6, CH	36.9, CH	44.3, CH	36.7, CH
21	20.1, CH_3	20.2, CH_3	20.4, CH_3	20.7, CH_3	20.3, CH_3	20.5, CH_3	20.2, CH_3	20.7, CH_3	20.7, CH_3	19.2, CH_3	21.2, CH_3	20.5, CH_3	18.9, CH_3	62.3, CH_2	19.2, CH_3
22	50.5, CH_2	50.5, CH_2	50.8, CH_2	50.8, CH_2	50.9, CH_2	50.1, CH_2	50.2, CH_2	45.2, CH_2	44.9, CH_2	31.5, CH_2	33.4, CH_2	33.3, CH_2	35.2, CH_2	31.1, CH_2	37.1, CH_2
23	209.6, C	209.7, C	210.1, C	209.9, C	209.8, C	209.8, C	209.5, C	67.5, CH	67.6, CH	36.2, CH_2	37.4, CH_2	37.3, CH_2	26.6, CH_2	25.6, CH_2	25.4, CH_2
24	47.6, CH_2	47.7, CH_2	47.8, CH_2	47.9, CH_2	47.9, CH_2	47.5, CH_2	47.5, CH_2	146.1, CH	145.7, CH	202.7, C	79.5, CH	76.7, CH	155.9, CH	125.9, CH	125.9, CH
25	36.1, CH	36.1, CH	36.1, CH	36.4, CH	36.4, CH	36.1, CH	36.0, CH	125.7, C	129.6, C	150.7, C	74.0, C	75.3, C	139.7, C	136.7, C	141.6, C
26	178.9, C	179.2, C	179.1, C	179.3, C	179.5, C	179.2, C	178.8, C	171.9, C	171.2, C	61.3, CH_2	67.8, CH_2	69.7, CH_2	196.6, C	68.6, CH_2	68.7, C
27	18.4, CH_3	18.3, CH_3	18.3, CH_3	18.4, CH_3	18.4, CH_3	18.4, CH_3	18.1, CH_3	14.7, CH_3	14.2, CH_3	123.5, CH_2	21.2, CH_3	21.0, CH_3	9.7, CH_3	14.4, CH_3	14.5, CH_3
28	68.9, CH_2	69.1, CH_2	67.2, CH_2	25.6, CH_3	67.4, CH_2	67.4, CH_2	12.1, CH_3	25.9, CH_3	20.7, CH_3	25.7, CH_3	25.8, CH_3	25.8, CH_3	25.5, CH_3	21.7, CH_3	16.3, CH_3
29	17.8, CH_3	17.3, CH_3	18.2, CH_3	21.8, CH_3	25.6, CH_3	17.9, CH_3	–	21.8, CH_3	27.9, CH_3	25.5, CH_3	25.5, CH_3	25.6, CH_3	25.8, CH_3	25.8, CH_3	28.8, CH_3
30	24.6, CH_3	24.5, CH_3	25.6, CH_3	24.9, CH_3	25.0, CH_3	24.7, CH_3	24.7, CH_3	25.4, CH_3	24.6, CH_3	21.7, CH_3	18.1, CH_3	18.1, CH_3	21.9, CH_3	25.6, CH_3	25.9, CH_3
$CH_3\underline{CO}$	–	170.9, C	–	–	–	170.8, C	–	–	–	–	–	–	–	–	–
$\underline{CH_3}CO$	–	21.0, CH_3	–	–	–	20.9, CH_3	–	–	–	–	–	–	–	–	–

[a] Measured in $CDCl_3$.
[b] Measured in C_5D_5N. The assignments were based on HSQC and HMBC experiments.

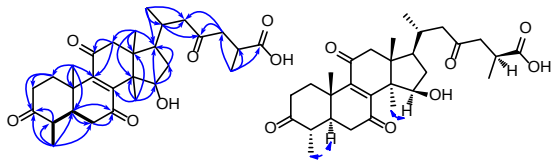


Fig. 4. Key HMBC (H → C), ^{1}H-^{1}H COSY (H—H) and ROESY (H···H) correlations of **7**.

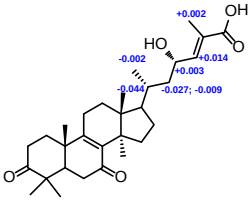
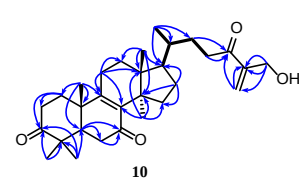


Fig. 5. Values of δ_S–δ_R of the MTPA esters of **8**.

(1706 cm^{-1}) groups were deduced by analysis of the IR spectrum. Its UV spectrum also indicated α,β-unsaturated carbonyl groups (λ_{max} 252 nm). Analysis of the 1D-NMR spectroscopic data (Tables 1 and 2) of **9** suggested it had the same side-chain moiety (23-hydroxy-24-en-26-oic acid) as in **8**.

In the ^{13}C-DEPT NMR spectra of **9**, signals at δ 203.7, 151.8, 150.7 and 202.6 in **9** suggested presence of a conjugated system (C-7/C-8/C-9/C-11). Meanwhile, comparison of its 1D-NMR spectroscopic data with compound **8** indicated presence of an oxymethine in **9** instead of a methylene at C-15 in **8**. This was confirmed by HMBC correlations of H_3-30, H_2-16 and H-17 with the oxymethine (δ 75.3), together with ^{1}H-^{1}H COSY correlations of H-15/H_2-16/H-17. Based on the ROESY correlation of H_3-30/H-15, the relative configuration of 15-OH was β. Compound **9** was accordingly determined to be 15β,23*S*-dihydroxy-3,7,11-trioxo-5α-lanost-8,24*E*-dien-26-oic acid, named hainanic acid B.

Compound **10** had a molecular formula of $C_{30}H_{44}O_4$, as determined by HREIMS (*m/z* 468.3246 [M]$^+$, calcd 468.3240). Its UV absorption band at 252 nm and IR bands at 1708 and 1629 cm^{-1} indicated the presence of a ketone carbonyl and a double bond. Its 1D spectroscopic NMR data suggested it had the same four-ring substructure as ganoderone A (Niedermeyer et al., 2005). In its ^{13}C NMR spectra, signals at δ 214.5, δ 198.2, δ 139.7 and δ 163.4 were attributed to the ketone carbonyl group at C-3 and an α,β-unsaturated carbonyl group at C-7/C-8/C-9, respectively, which were also supported by analysis of the 2D NMR spectra. Apart from that, the ^{13}C-DEPT NMR spectra also showed presence of a ketone carbonyl, a terminal double bond and an oxymethylene in the side-chain of **10**.



10

Fig. 6. Key HMBC (H → C) and ^{1}H-^{1}H COSY (H—H) correlations of **10**.

One doublet methyl proton resonance at δ 0.98 (d, J = 6.0 Hz) was assigned as Me-21 from HMBC correlations (Fig. 6) of H_3-21 with C-17, C-20 and C-22. Protons of the sp^2 methylene also gave an HMBC correlation with the oxymethylene group. Thus, the terminal double bond and oxymethylene were linked to C-26, C-25,

and C-27, respectively. Meanwhile, both H_2-26 and H_2-27 were correlated with the ketone carbonyl in the HMBC spectrum of **10**, suggesting that it was connected to C-24. Consequently, structure **10** was established to be 3,7,24-trioxo-5α-lanost-8,25-dien-26-ol.

A molecular formula of $C_{30}H_{48}O_5$ was established for compound **11** from its HREIMS (*m/z* 488.3508 $[M]^+$, calcd 488.3502). Its IR spectrum showed presence of hydroxyl (3433 cm^{-1}) and α,β-unsaturated carbonyl (1709 cm^{-1}) groups. Its ^{13}C NMR spectra (C_5D_5N, 150 MHz) implied that its four-ring substructure was almost the same as **10**; nevertheless, three characteristic oxygenated carbons (δ 69.7, CH_2; δ 76.7, CH; δ 75.3, C) suggested it had the same side-chain moiety as for ganodermanontriol. Furthermore, HMBC correlations of H_2-22, H_2-23 with C-24 (δ 76.7), of H_2-23, H-24, H_3-26 and H_2-27 with C-25 (δ 75.3) and of H-26 and H-24 with C-27 (δ 69.7) confirmed that C-24, C-25 and C-27 of **11** were oxygenated.

Kennedy et al. (2011) synthesized four side-chain triol diastereomers and those ^{13}C-NMR spectroscopic data at C-24, C-25, and C-26 in the four side-chain triol diastereomers were obviously distinct. The absolute configuration of C-24 and C-25 was, however, determined by comparing the chemical shifts of C-24, C-25 and C-26 in **11** with those of the triol diastereomers. In the ^{13}C NMR spectra ($CDCl_3$, 150 MHz) of **11**, resonances at δ 79.5, δ 74.0 and δ 67.8 (C-24, C-25 and C-26) were the same as those of ganodermanontriol, which established C-24 and C-25 to be *S* and *R*, respectively. Thus, structure **11** was consequently established as 24*S*,25*R*-dihydroxy-3,7-dioxo-5α-lanost-8-en-26-ol.

The molecular formula $C_{30}H_{44}O_4$ of **12** was determined by HREIMS (found *m/z* 468.3247, calcd 468.3240 for $[M]^+$). Its IR spectrum indicated presence of hydroxyl (3439 cm^{-1}) and α,β-unsaturated carbonyl (1709 cm^{-1}) functional groups. Its ^1H NMR spectrum showed six singlet methyls at δ 1.82, 1.54, 1.43, 1.10, 1.04 and 0.82, one doublet methyl at δ 1.00 (*J* = 6.0 Hz), one oxygenated methine at δ 4.81 (m), one sp^2 methine at δ 6.51 (t, *J* = 7.1 Hz) and one aldehyde proton at δ 9.59 (s), respectively. In its ^{13}C NMR spectra, seven methyls, eight methylenes, six methines (one oxygenated, one olefinic and one aldehyde) and eight quaternary carbons (two ketones, three olefinic carbons) were also observed. Its 1D-NMR spectroscopic data were similar to those of lucialdehyde D (Niedermeyer et al., 2005), except for an oxygenated methine replacing the ketone carbonyl group at C-11 in lucialdehyde D. This was confirmed by the HMBC correlations of H_2-12 with C-11 and of H-11 with C-9, C-8, C-10, C-12 and C-13, along with the ^1H-^1H COSY correlation of H-11/H_2-12.

On the basis of the ROESY experiment, a correlation of H-11/H_3-19 assigned the 11-OH group as being α-oriented. The absence of a ROESY correlation of H-24/H_3-27 suggested that the configuration of the double bond at C-24 was *E*. Therefore, structure **12** was 11α-hydroxy-3,7-dioxo-5α-lanost-8,24*E*-dien-26-al, named hainanaldehyde A.

The 1D-NMR spectroscopic data of **13** resembled those of ganoderone A (**16**) (Niedermeyer et al., 2005). Its HREIMS at *m/z* 470.3383 $[M]^+$ indicated that it had a molecular formula of $C_{30}H_{46}O_4$ (calcd 470.3396), with one oxygen more than ganoderone A. Comparison of the ^1H-NMR spectroscopic data between these two compounds showed that a doublet methyl signal at δ 0.93 (d, *J* = 5.5 Hz, H_3-21) in ganoderone A was replaced by oxymethylene resonances (each at δ 3.99, dd, *J* = 4.9 and 10.8 Hz; δ 4.11, dd, *J* = 3.3 and 10.8 Hz) in **13**. Observed HMBC correlations of H-17, H-21 and H-22 with C-21, and of H-21 with C-20 and C-17, together with ^1H-^1H COSY correlations of H-17/H_2-20/H_2-21/H_2-22/H_2-23/H-24, confirmed that a hydroxyl group was linked to C-21 in **13**. The *E*-configuration of C-24 was determined by analysis of the ROESY correlation of H-24/H_2-26. Compound **13** was consequently assigned as 21-hydroxy-3,7-dioxo-5α-lanost-8,24*E*-dien-26-ol.

Table 3
Cytotoxicity data of compounds **10**, **12** and **17** (IC_{50}, μM).

Compound	HL-60	SMMC-7721	A-549	MCF-7	SW480
10	15.70	15.52	15.81	20.08	>40
12	15.50	15.51	22.54	19.91	>40
17	37.47	36.53	36.92	24.55	>40
cis-Dichlorodiamine-platinum(II) (DDP)[a]	1.05	6.76	6.01	15.38	16.31

[a] DDP as the positive control.

Compound **14** had a molecular formula of $C_{30}H_{48}O_4$ (calcd for $C_{30}H_{48}O_4$, *m/z* 472.3553) as indicated from its HREIMS at *m/z* 472.3542 $[M]^+$. Its 1D-NMR spectroscopic data (Tables 1 and 2) indicated it was similar to lucidadiol (Gonzalez et al., 1999), with differences suggesting an oxygenated methine (δ 3.49, m) in **14** replaced the methylene in lucidadiol. This oxymethine group was established to be C-11 by HMBC correlations of H-5 and H_2-6 with the carbonyl carbon (δ 200.9) and of H_2-12 and the oxymethine (δ 77.9). Its ROESY spectrum gave correlations of H-3/H-5, of H-11/H_3-19 which demonstrated that the relative configuration of H-3 and H-11 were α and β, respectively. 24*E* was also determined by the ROESY correlation of H-24/H_2-26. Hence, structure **14** was determined as 3β,11α-dihydroxy-7-oxo-5α-lanost-8,24*E*-dien-26-ol.

Ganoderma acids possessing a 23-oxo-5α-lanost-8-en-26-oic acid skeleton are characteristic chemical constituents of *Ganoderma*. The absolute configuration of C-25 in ganoderic acid C from *G. lucidum* was shown through an X-ray structure to be *R* (Hirotani et al., 1985); nevertheless, the X-ray crystallographic analysis of **1a** had a 25*S* configuration. Meanwhile, the relative configuration of C-15 in ganoderma acids from this fungus was determined to be β by the ROESY experiments, while ganoderma acids previously reported in the literature were α (Kikuchi et al., 1985a,b, 1986). A structure–activity relationship for inhibition of 5α-reductase and aldose reductase by triterpenoids suggested the carboxyl group at C-26 was a key active group (Liu et al., 2006; Fatmawati et al., 2011b). Thus, the configurations of C-15 and C-25 should be considered when studying bioactivities of ganoderma acids.

2.2. X-ray crystal analysis of **1a**

Its structure was solved by direct methods using the program SHELXS-97 (Sheldrich, 1996, University of Gottingen; Gottingen, Germany), then refined by SHELXS, with refinement on F^2 against all reflections. All esd's are estimated using the full covariance matrix. Non-hydrogen atoms were refined anisotropically. Hydrogen atoms were located by geometry and riding on the related atoms during refinements with a temperature factor of 1.2 or 1.5 times the later.

2.3. Cytotoxicities of isolates

Compounds **1**, **3**–**8**, **10**–**14**, **16**–**19** were tested for their cytotoxicities against HL-60, SMMC-7721, A-549, MCF-7 and SW480 cells using the MTT method, with DDP (*cis*-dichlorodiamineplatinum II) used as positive control. Compounds **10**, **12** and **17** showed selective inhibitory activities against HL-60, SMMC-7721, A-549 and MCF-7, with IC_{50} values in the range of 15–40 μM, whereas they did not display any cytotoxicity against SW480 cells (Table 3).

3. Conclusion

In this contribution, compounds **1**–**19** containing ganoderma acids and ganoderma alcohols, were isolated from fruiting bodies of *G. hainanense*. Among them, compound **7** with a 29-norlanostane skeleton was identified from the genus *Ganoderma* for the first

time. Interestingly, configurations of C-15 and C-25 in ganoderma acids from *G. hainanense* were different from those of ganoderma acids from other *Ganoderma* species. Considering that *G. hainanense* possess abundant lanostane triterpenoids, this fungus might also exhibit broad spectrum of biological activities, similar to *G. lucidum* and *G. sinense*.

4. Experimental

4.1. General experimental procedures

Optical rotations were obtained with a Jasco P-1020 polarimeter. ^{1}H and ^{13}C NMR spectra were measured on Bruker AV-400 and DRX-500 instruments (Bruker, Zurich, Switzerland) using TMS as internal standard for chemical shifts. Chemical shifts (δ) were expressed in *ppm* with reference to the TMS resonance. ESIMS and HRTOF-ESIMS data were acquired on an API QSTAR Pulsar spectrometer. Infrared spectra were recorded on a Bruker Tensor-27 instrument by using KBr pellets. An Agilent 1100 series instrument equipped with an Agilent ZORBAX SB-C18 column (5 μm, 9.6 mm × 250 mm) was used for high-performance liquid chromatography (HPLC) analysis.

4.2. Fungal materials

Fruiting bodies of *G. hainanense* were purchased in July 2012 from the Juhuacun Traditional Chinese Medicine Market in Kunming. The mushroom was identified by Prof. Liu Peigui, a fungal taxonomist who works at the Kunming Institute of Botany, Chinese Academy of Science. A specimen (No. 12071001) is deposited in the Herbarium of the Department of Taxonomy, Kunming Institute of Botany, Chinese Academy of Sciences.

4.3. Extraction and isolation

Fruiting bodies of *G. hainanense* (1.5 kg) were powdered and exhaustively percolated with acetone at room temperature. The combined extracts were concentrated to yield a residue (150 g), which was subjected to over HP-20 macroreticular resin column chromatography (CC), eluting with MeOH-H_2O (50:50; 70:30; 90:10 and 100:0), to yield three fractions: Fr. 1 (3.0 g) from MeOH-H_2O (50:50), Fr. 2 (3.2 g) from MeOH-H_2O (70:30) and Fr. 3 (90%, 13 g) from MeOH-H_2O (90:10). Fr. 3 (13 g) was further separated by elution silica gel CC (200–300 mesh), eluting with $CHCl_3$–MeOH (100:0, 85:1, 50:1, 20:1 and 5:1, stepwise) to afforded four sub-fractions (Fr.3-1, Fr.3-2, Fr.3-3 and Fr.3-4). Fr.3-1 (85:1) was recrystallized to give ganoderone A **16** (50 mg), from MeOH. Fr.3-2 (50:1) was separated by silica gel CC ($CHCl_3$–acetone) and further purified by semi-preparative HPLC (CH_3CN–H_2O gradient elution) to obtain compounds **10** (6.1 mg), **12** (4.9 mg), lucidadiol (43 mg, **17**) and 4,4,14α-trimethyl-3,7-dioxo-5α-chol-8-en-24-oic acid (8.4 mg, **19**), respectively. Fr.3-3 (20:1) was subjected to P-TLC (Preparative Thin-Layer Chromatography) to afford compound **1** (5.2 mg).

Fr.3-4 (5:1) and Fr.2 were combined and subjected to silica gel CC ($CHCl_3$–MeOH, 80:1, 50:1, 20:1 and 5:1) and monitored by TLC to obtain four fractions (Fr.2-1, Fr.2-2, Fr.2-3 and Fr.2-4). Each fraction was separated on a reversed-phase C18 column (MeOH–H_2O, 50:50 → 70:30). Fr.2-1 was divided into four subfractions (Fr.2-1-1 → Fr.2-1-4), namely MeOH-H_2O (50:50; 55:45; 60:40 and 70:30, v/v). Compounds **2** (8.3 mg) and **7** (10.1 mg) were obtained from Fr.2-1-1 by P-TLC. Fr.2–1-2 was further purified by semi-preparative HPLC (CH_3CN–H_2O, 48:52 → 60:40, v/v, gradient elution, 30 min) to afford **14** (6.7 mg, 26.00 min). Fr.2-1-3 was subjected to P-TLC to yield ganodermanontriol (2.5 mg, **18**). Fr2-1-4 was separated by semi-preparative HPLC (CH_3CN–H_2O, 50:50 → 26:74, gradient elution, 32 min, 60%, 4 min) to produce **3** (4.7 mg, 19.26 min), **4** (3.2 mg, 24.11 min), **6** (4.3 mg, 20.69 min), **8** (4.4 mg, 25.15 min), **11** (5.2 mg, 22.78 min), **13** (3.2 mg, 24.53 min) and ganoderol J (2.1 mg, 30.34 min, **15**), respectively.

Fr.2–2 was divided into four sub-fractions (Fr.2-2-1 → Fr.2-2-4) by RP-18 (MeOH–H_2O, 45:55; 50:50; 55:45 and 60:40, v/v). Fr.2-2-2 was purified by semi-preparative HPLC (CH_3CN–H_2O, 35:65 → 40:60, gradient elution, 20 min) to obtain **5** (3.8 mg, 11.02 min). Fr.2-2-3 was treated by P-TLC to yield **9** (2.1 mg).

4.3.1. Ganohainanic acid A (*1*)

White needles melting point 190–191 °C; $[\alpha]_D^{18}$ +88.34 (*c* 0.90, MeOH); UV (MeOH) λ_{max} (log ε) nm: 266 (4.29), 227 (4.39) nm; IR (KBr) ν_{max}: 3423, 2975, 2953, 2925, 2874, 2854, 1698, 1692, 1658, 1617, 1462, 1424, 1396, 1380, 1315, 1241 and 1173 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 529 [M−H]$^-$, HREIMS *m/z* 530.2885 [M]$^+$ (calcd for $C_{30}H_{42}O_8$, 530.2880).

4.3.2. Crystal Data of *1a*

$C_{31}H_{44}O_8{\cdot}H_2O$, M = 562.68, orthorhombic, space group $P2_12_12$, a = 13.7003 (6) Å, b = 29.7316 (13) Å, c = 7.1476 (3) Å, $\alpha = \beta = \gamma = 90°$, V = 2911.4 (2) Å^3, Z = 4, d = 1.284 mg/cm^3, crystal dimensions 1.10 × 0.38 × 0.07 mm was used for measurements on a Bruker APEX DUO diffractometer with a graphite monochromator (Φ/ω scans, $2\theta_{max}$ = 69.43°), Cu K_α radiation. The total number of independent reflections measured was 24,865, of which 5183 were observed ($|F|^2 \geqslant 2\sigma|F|^2$). Final indices: R_1 = 0.0495, wR_2 = 0.1434 ($w = 1/\sigma|F|^2$), S = 1.057. The crystal structure of **1a** reported here is deposited with the Cambridge Crystallographic Data Centre (deposition number: 970291). Copies of these data can be obtained free of charge via the Internet at www.ccdc.cam.ac.uk/conts/retrieving.html (or from the Cambridge Crystallographic Data Center, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax (+44) 1223-336-033; or e-mail: deposit @ccdc.cam.ac.uk).

4.3.3. Acetyl ganohainanic acid A (*2*)

White powder, $[\alpha]_D^{20}$ +28.33 (*c* 0.20, MeOH); UV (MeOH) λ_{max} (log ε) nm: 262 (3.98), 203 (3.89) nm; IR (KBr) ν_{max}: 3442, 2927, 1742, 1710, 1672, 1658, 1626, 1582, 1461, 1453, 1425, 1380, 1317, 1233, 1202 and 1165 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 571 [M−H]$^-$, HREIMS *m/z* 572.2978 [M]$^+$ (calcd for $C_{32}H_{44}O_9$, 572.2985).

4.3.4. Ganohainanic acid B (*3*)

White powder, $[\alpha]_D^{18}$ +29.5 (*c* 0.25, MeOH); UV (MeOH) λ_{max} (log ε) nm: 252 (3.86), 202 (3.72) nm; IR (KBr) ν_{max}: 3440, 2968, 1930, 1708, 1584, 1460, 1416, 1377, 1280, 1269, 1237, 1216, 1152 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 499 [M−H]$^-$, HREIMS *m/z* 500.3121 [M]$^+$ (calcd for $C_{30}H_{44}O_6$, 500.3138).

4.3.5. Ganohainanic acid C (*4*)

White powder, $[\alpha]_D^{20}$ +8.3 (*c* 0.32, MeOH); UV (MeOH) λ_{max} (log ε) nm: 256 (3.82), 201 (3.60) nm; IR (KBr) ν_{max}: 3441, 2969, 2934, 2881, 1709, 1672, 1641, 1580, 1461, 1413, 1379, 1346, 1315, 1238, 1178, 1116, 1079 and 1033 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 499 [M−H]$^-$, HREIMS *m/z* 500.3140 [M]$^+$ (calcd for $C_{30}H_{44}O_6$, 500.3138).

4.3.6. Ganohainanic acid C (*5*)

White powder, $[\alpha]_D^{17}$ −29.9 (*c* 0.14, MeOH); UV (MeOH) λ_{max} (log ε) nm: 254 (3.86), 198 (3.54) nm; IR (KBr) ν_{max}: 3439, 2956, 2928, 1703, 1632, 1572, 1566, 1453, 1425, 1383, 1317, 1271, 1236,

1201 and 1109 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 539 $[M+Na]^+$, HREIMS *m/z* 516.3039 $[M]^+$ (calcd for $C_{30}H_{44}O_7$, 516.3087).

*4.3.7. Acetyl ganohainanic acid C (**6**)*

White powder, $[\alpha]_D^{20}$ −25.7 (*c* 0.23, MeOH); UV (MeOH) λ_{max} (log ε) nm: 254 (3.86), 200 (3.62) nm; IR (KBr) ν_{max}: 3432, 2952, 2925, 2854, 1740, 1710, 1639, 1468, 1381, 1317, 1237, 1178, 1124 and 1107 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 557 $[M-H]^-$, HREIMS *m/z* 558.3171 $[M]^+$ (calcd for $C_{32}H_{46}O_8$, 558.3193).

*4.3.8. Ganohainanic acid E (**7**)*

White needles, $[\alpha]_D^{21}$ +66.61 (*c* 0.12, MeOH); UV (MeOH) λ_{max} (log ε) nm: 264 (3.80), 201 (3.74) nm; IR (KBr) ν_{max}: 3441, 2971, 2937, 2883, 1711, 1671, 1462, 1424, 1415, 1379 and 1237 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 499 $[M-H]^-$, HREIMS *m/z* 500.2769 $[M]^+$ (calcd for $C_{29}H_{40}O_7$, 500.2774).

*4.3.9. Hainanic acid A (**8**)*

White powder, $[\alpha]_D^{19}$ +22.6 (*c* 0.27, MeOH); UV (MeOH) λ_{max} (log ε) nm: 252 (3.80), 203 (3.89) nm; IR (KBr) ν_{max}: 3440, 2968, 2926, 2890, 2880, 1709, 1691, 1640, 1461, 1452, 1423, 1415, 1382, 1345, 1272, 1239 and 1116 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 483 $[M-H]^-$, HREIMS *m/z* 484.3183 $[M]^+$ (calcd for $C_{30}H_{44}O_5$, 484.3189).

*4.3.10. Preparation of (R)- and (S)-MTPA ester derivatives of **8** by the Mosher ester procedure*

Two portions (each 1.0 mg) of **8** were individually treated with (*R*)-(−)- and (*S*)-(+)-α-methoxy-α-(trifluoromethyl)phenylacetyl chloride (1 μl) in deuterated pyridine (20 μl) directly in separate NMR tubes, with each NMR tube carefully shaken to mix the sample and MTPA chloride at room temperature, to afford the (*R*)- and (*S*)-MTPA ester derivatives (**8r** and **8s**) after 24 h. The ^{1}H NMR spectroscopic data of the (*R*)-MTPA ester derivative (**8r**) of **8** (120 MHz, pyridine-d_5, data was obtained directly from the reaction NMR tube directly and assigned on the basis of correlations of the ^{1}H-^{1}H COSY spectrum): δ 1.704 (1H, m, H-20), δ 1.225 (3H, d, *J* = 6.3 Hz, Me-21), δ 1.876 (1H, m, H-22α), δ 2.121 (1H, m, H-22β), δ 5.083 (1H, m, H-23), δ 7.473 (1H, m, H-24), δ 2.231 (3H, s, Me-27). In a similar way, the ^{1}H NMR spectroscopic data of the (*S*)-MTPA ester derivative (**8s**) of **8** was: δ 1.696 (1H, m, H-20), δ 1.227 (3H, d, *J* = 6.3 Hz, Me-21), δ 1.867 (1H, m, H-22α), δ 2.094 (1H, m, H-22β), δ 5.086 (1H, m, H-23), δ 7.487 (1H, m, H-24), δ 2.233 (3H, s, Me-27).

*4.3.11. Hainanic acid B (**9**)*

White powder, $[\alpha]_D^{20}$ +149.2 (*c* 0.12, MeOH); UV (MeOH) λ_{max} (log ε) nm: 252 (3.85), 203 (3.73) nm; IR (KBr) ν_{max}: 3440, 2965, 2930, 1706, 1675, 1660, 1549, 1510, 1451, 1424, 1377, 1249, 1233, 1156, and 1114 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 513 $[M-H]^-$, HREIMS *m/z* 514.2931 $[M]^+$ (calcd for $C_{30}H_{42}O_7$, 514.2964).

*4.3.12. 3,7,24-trioxo-8,25-dien-5α-lanosta-26-ol (**10**)*

White powder, $[\alpha]_D^{19}$ −162.72 (*c* 0.09, MeOH); UV (MeOH) λ_{max} (log ε) nm: 252 (3.69), 202 (3.70) nm; IR (KBr) ν_{max}: 3432, 2959, 2924, 2854, 1708, 1629, 1426, 1384, 1104 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT data, see Tables 1 and 2; ESIMS *m/z* 491 $[M+Na]^+$, HREIMS *m/z* 468.3246 $[M]^+$ (calcd for $C_{30}H_{44}O_4$, 468.3240).

*4.3.13. 24S,25R-dihydroxy-3,7-dioxo-8-en-5α-lanosta-26-ol (**11**)*

White powder, $[\alpha]_D^{20}$ +8.6 (*c* 0.23, MeOH); UV (MeOH) λ_{max} (log ε) nm: 254 (3.83), 201 (3.59) nm; IR (KBr) ν_{max}: 3433, 2969, 2928, 2881, 1709, 1639, 1460, 1416, 1380, 1270, 1239, 1115, 1073, 1066 and 1044 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 487 $[M-H]^-$, HREIMS *m/z* 488.3508 $[M]^+$ (calcd for $C_{30}H_{48}O_5$, 488.3502).

*4.3.14. Hainanaldehyde A (**12**)*

White powder, $[\alpha]_D^{20}$ −31.11 (*c* 0.06, MeOH); UV (MeOH) λ_{max} (log ε) nm: 232 (4.26), 195 (3.96) nm; IR (KBr) ν_{max}: 3439, 2957, 2924, 2854, 1708, 1636, 1458, 1452, 1424, 1382, 1175, 1114 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 491 $[M+Na]^+$, HREIMS *m/z* 468.3247 $[M]^+$ (calcd for $C_{30}H_{44}O_4$, 468.3240).

*4.3.15. 21-hydroxy-3,7-dioxo-8,24E-dien-5α-lanosta-26-ol (**13**)*

White powder, $[\alpha]_D^{20}$ +3.2 (*c* 0.22, MeOH); UV (MeOH) λ_{max} (log ε) nm: 256 (3.83), 204 (3.48) nm; IR (KBr) ν_{max}: 3442, 2977, 2967, 2877, 1706, 1652, 1567, 1455, 1356, 1379, 1343 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 469 $[M-H]^-$, HREIMS *m/z* 470.3383 $[M]^+$ (calcd for $C_{30}H_{46}O_4$, 470.3396).

*4.3.16. 3β,7β-dihydroxy-11-oxo-8,24E-dien-5α-lanosta-26-ol (**14**)*

White powder, $[\alpha]_D^{16}$ +11.3 (*c* 0.14, MeOH); UV (MeOH) λ_{max} (log ε) nm: 251 (3.73), 201 (3.77) nm; IR (KBr) ν_{max}: 3430, 2962, 2930, 1657, 1631, 1565, 1549, 1510, 1451, 1424, 1377, 1249, 1233, 1156, and 1114 cm^{-1}; For ^{1}H-NMR and ^{13}C-DEPT spectroscopic data, see Tables 1 and 2; ESIMS *m/z* 471 $[M-H]^-$, HREIMS *m/z* 472.3542 $[M]^+$ (calcd for $C_{30}H_{48}O_4$, 4723553).

4.4. Cytotoxicity assays

Cytotoxicities of compounds **1**, **3**–**8**, **10**–**14**, **16**–**19** against HL-60, SMMC-7221, A-549, MCF-7 and SW480 cells were measured using methyl thiazole tetrazolium (MTT) assay (Alley et al., 1988). IC_{50} values were determined as concentration of a compound resulting in cell growth inhibition by 50%. All samples were assayed in triplicate.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.phytochem.2014.10.009.

References

Akihisa, T., Nakamura, Y., Tagata, M., Tokuda, H., Yasukawa, K., Uchiyama, E., Suzuki, T., Kimura, Y., 2007. Anti-inflammatory and anti-tumor-promoting effects of triterpene acids and sterols from the fungus *Ganoderma lucidum*. Chem. Biodivers. 4, 224–231.

Alley, M.C., Scudiero, D.A., Monks, A., Hursey, M.L., Czerwinski, M.J., Fine, D.L., Abbott, B.J., Mayo, J.G., 1988. Feasibility of drug screening with panels of human tumor cell lines using a microculture tetrazolium assay. Cancer Res. 48, 589–601.

Cheng, C.R., Yue, Q.X., Wu, Z.Y., Song, X.Y., Tiao, S.J., Wu, X.H., Xu, P.P., Liu, X., Guan, S.H., Guo, D.A., 2010. Cytotoxic triterpenoids from *Ganoderma lucidum*. Phytochemistry 71, 1579–1585.

Fatmawati, S., Shimizu, K., Kondo, R., 2011a. Ganoderol B: a potent α-glucosidase inhibitor isolated from the fruiting body of *Ganoderma lucidum*. Phytomedicine 18, 1053–1055.

Fatmawati, S., Shimizu, K., Kondo, R., 2011b. Structure–activity relationships of ganoderma acids from *Ganoderma lucidum* as aldose reductase inhibitors. Bioorg. Med. Chem. Lett. 21, 7295–7297.

Gonzalez, A.G., Leon, F., Rivera, A., Munoz, C.M., Bermejo, J., 1999. Lanostanoid triterpenes from *Ganoderma lucidum*. J. Nat. Prod. 62, 1700–1701.

Grienke, U., Mihaly-Biaon, J., Schuster, D., Afonyushkin, T., Binder, M., Guan, S.H., Cheng, C.R., Wolber, G., Stuppner, H., Guo, D.A., Bochkov, V.N., Rollinger, J.M., 2011. Pharmacophore-based discovery of FXR-agonists. Part II: Identification of bioactive triterpenes from *Ganoderma lucidum*. Bioorg. Med. Chem. 19, 6779–6791.

Hirotani, M., Furuya, T., Shiro, M., 1985. A ganoderic acid derivtive, a highly oxygenated lanostane-type triterpenoids from *Ganoderma lucidum*. Phytochemistry 24, 2055–2061.

Kennedy, E.M., P'Pool, S.J., Jiang, J.H., Sliva, D., Minto, R.E., 2011. Semisynthesis and biological evaluation of ganodermanontriol and its stereoisomeric triols. J. Nat. Prod. 74, 2332–2337.

Kikuchi, T., Matsuda, S., Kadota, S., Murai, Y., 1985a. Ganoderic acid D, E, F, and H and lucidenic acid D, E, and F, new triterpenoids from *Ganoderma lucidum*. Chem. Pharm. Bull. 33, 2624–2627.

Kikuchi, T., Matsuda, S., Murai, Y., Ogita, Z., 1985b. Ganoderic acid G and I and ganolucidic acid A and B, new triterpenoids from *Ganoderma lucidum*. Chem. Pharm. Bull. 33, 2628–2631.

Kikuchi, T., Kanomi, S., Murai, Y., Kadota, S., Tsubono, K., Ogita, Z.I., 1986. Constituents of the fungus *Ganoderma lucidum* (FR.) Karst. III. Structures of ganolucidic acids A and B, new lanostane-type triterpenoids. Chem. Pharm. Bull. 34, 4030–4036.

Li, B.Y., Wu, X.L., Song, B., Li, T.H., Zhong, J.X., Chen, H.Q., 2013. Distribution of the medicinal *Ganoderma* ssp. in Hainan. Guangdong Agric. Sci. 3, 171–172.

Liu, J., Kurashiki, K., Shimizu, K., Kondo, R., 2006. Structure-active relationship for inhibition of 5a-reductase by triterpenoids isolated from *Ganoderma lucidum*. Bioorg. Med. Chem. 24, 8654–8660.

Liu, J.Q., Wang, C.F., Li, Y., Luo, H.R., Qiu, M.H., 2012a. Isolation and bioactivity evaluation of terpenoids from the medicinal fungus *Ganoderma sinense*. Planta Med. 78, 368–376.

Liu, J., Kurashiki, K., Fukuta, A., Kaneko, S., Suimi, Y., Shimizu, K., Kondo, R., 2012b. Analytical methods quantitative determination of the representative triterpenoids in the extracts of *Ganoderma lucidum* with different growth stages using high-performance liquid chromatography for evaluation of the 5a-reductase inhibitory properties. Food Chem. 133, 1034–1038.

Ma, Q.Y., Luo, Y., Huang, S.Z., Guo, Z.K., Dai, H.F., Zhao, Y.X., 2013. Lanostane triterpenoids with cytotoxic activities from the fruiting bodies of *Ganoderma hainanense*. J. Asian Nat. Prod. Res. 15, 1214–1219.

Min, B.S., Nakamura, N., Miyashiro, H., Bae, K.W., Hattori, M., 1998. Triterpenoids from the spores of *Ganoderma lucidum* and their inhibition activity against anti-Hiv-1 protease. Chem. Pharm. Bull. 46, 1607–1612.

Min, B.S., Gao, J.J., Nakamura, N., Hattori, M., 2000. Triterpenes from the spores of *Ganoderma lucidum* and their cytotoxicity against meth-A and LLC tumor cells. Chem. Pharm. Bull. 48, 1026–1033.

Niedermeyer, T.H.J., Lindequist, U., Mentel, R., Gordes, D., Schmidt, E., Thurow, K., Lark, M., 2005. Antiviral terpenoid constituents of *Ganoderma pfeifferi*. J. Nat. Prod. 68, 1728–1731.

Nishitoba, T., Sato, H., Sakamura, S., 1985. New terpenoids, ganoderic acid J and ganolucidic acid C, from the fungus *Ganoderma lucidum*. Agric. Biol. Chem. 49, 3637–3638.

Nishitoba, T., Sato, H., Sakamura, S., 1986. New terpenoids ganolucidic acid D, ganderic acid L, lucidone C and lucidenic acid G, from the fungus *Ganoderma lucidum*. Agric. Biol. Chem. 50, 809–911.

Nishitoba, T., Oda, K., Sato, H., Sakamura, S., 1988. Novel triterpenoids from the fungus *Ganoderma lucidum*. Agric. Biol. Chem. 52, 367–372.

Niu, X.M., Qiu, M.H., Li, Z.R., Lu, Y., Cao, P., Zheng, Q.T., 2004. Two novel 3,4-*seco*-trinorlanostane triterpenoids isolated from *Ganoderma fornicatum*. Tetrahedron Lett. 45, 2989–2993.

Peng, X.R., Liu, J.Q., Han, Z.H., Yuan, X.X., Luo, H.R., Qiu, M.H., 2013. Protective effects of triterpenoids from *Ganoderma resinaceum* on H_2O_2-induced toxicity in HepG2 cells. Food Chem. 141, 920–926.

Sanodiya, B.S., Thakur, G.S., Baqhel, R.K., Prasad, G.B., Bisen, P.S., 2009. *Ganoderma lucidum*: a potent pharmacological macrofungus. Curr. Pharm. Biotechnol. 10, 717–742.

Sheldrick, G.M., 1996. SADABS: Program for Empirical Absorption Correction of Area Detector Data. University of Göttingen, Göttingen, Germany.

Su, B.N., Park, E.J., Mbwambo, Z.H., Santarsiero, B.D., Mesecar, A.D., Fong, H.H.S., Pezzuto, J.M., Kinghorn, A.D., 2002. New chemical constituents of *Euphorbia quinquecostata* and absolute configuration assignment by a convenient Mosher ester procedure carried out in NMR tubes. J. Nat. Prod. 65, 1278–1282.

Wang, C.F., Liu, J.Q., Yan, Y.X., Chen, J.C., Lu, Y., Guo, Y.H., Qiu, M.H., 2010. Three new triterpenoids containing four-membered ring from the fruiting body of *Ganoderma sinense*. Org. Lett. 12, 1656–1659.

Wu, T.S., Shi, L.S., Kuo, S.C., 2001. Cytotoxicity of *Ganoderma lucidum* triterpenes. J. Nat. Prod. 64, 1121–1122.

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